

Nina Gruteser

609-285-7622 | ngrutese@purdue.edu | [linkedin.com/in/nina-gruteser](https://www.linkedin.com/in/nina-gruteser) | github.com/njsumi

EDUCATION

Purdue University (GPA: 3.86/4.00)

West Lafayette, IN

Bachelor of Science in Computer Science Honors and Artificial Intelligence

Aug. 2023 – May 2027

- **Awards:** College of Science Dean's List & Semester Honors (Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025), NSF Student Travel Grant Award (IEEE/ACM CHASE 2022)
- **Relevant Courses:** Analysis of Algorithms, Introduction to AI, Data Mining & Machine Learning, Systems Programming, Competitive Programming, Data Structures & Algorithms, Computer Architecture, C Programming, Probability Theory

ETH Zürich

Zürich, Switzerland

Exchange Student, Computer Science (D-INFK)

Feb. 2026 – June 2026

- **Courses:** 3D Vision, Digital Humans, Cloud Computing Architecture

EXPERIENCE

Software Engineering Intern

Capital One | New York, NY

June 2026 – Aug. 2026

- Built new service component that provides a queryable interface for transaction status tracking by deploying data stores and implementing aggregations, allowing transaction initiators to self-serve the status of transactions.

Capital One | Richmond, VA

June 2025 – Aug. 2025

- Increased flexibility and efficiency of Apache Spark data pipelines by building a Scala abstraction layer that allows one executable to run dataset-dependent SQL transforms across multiple datasets used by 5,000+ ML models.
- Developed an automated validation tool to verify consistency between the previous and new pipeline's output, significantly reducing manual effort and improving reliability during migration.

Undergraduate Teaching Assistant

Aug. 2024 – May 2025

Purdue University

West Lafayette, IN

- Assisted 600+ students in Programming in C by leading office hours and lab sections, providing debugging assistance on online discussion forums, and helping students complete homework and prepare for exams.

Simons Research Fellow

June 2022 – Aug. 2022

SUNY Stony Brook

Stony Brook, NY

- Created a semi-automatic 3D segmentation model for head and neck tumors in PET Scans by pre-processing data and developing a CNN, which performs better than previous fully automated segmentation approaches [1]

PROJECTS

Reconstructing Human-Object Interactions | *Python, PyTorch, GVHMR, FoundationPose*

Feb. 2026 – June 2026

- Designed a hybrid pipeline that reconstructs 3D human and object poses from multi-view RGB-D videos.
- Combined monocular human motion recovery (GVHMR) with multi-view object tracking (FoundationPose), implemented coordinate space alignment and joint refinement, and evaluated the accuracy of the pipeline on the BEHAVE dataset.

Robotic VLM Benchmarking | *Python, Shell, Qwen2.5-VL, DeepSeek-VL2*

Feb. 2026 – June 2026

- Developed an evaluation pipeline to benchmark and model failures surrounding spatial reasoning and affordance understanding in open-source Vision-Language Models (VLMs) using the Robo2VLM dataset.
- Designed automated tests using Chain-of-Thought prompting, monocular depth maps via Depth Anything V2, and visual grounding sanity checks to uncover reasoning failures in robotic perception.

Cover Palette | *Python, Django, Spotify WebAPI, OpenAI API, HTML/CSS*

Sept. 2023 – Dec. 2023

- Developed a Generative AI playlist cover image generator, where users can obtain unique images for their playlists
- Implemented the Spotify WebAPI to extract song features from user playlists and engineered a dynamic prompt formula sent to DALL-E to generate and visualize unique cover art matching the playlist's musicality.

PUBLICATIONS

- [1] Nina Gruteser. "Head and Neck Tumor Segmentation With Sliced 3D PET Scans". IEEE/ACM CHASE 2022

SKILLS

Programming Languages: Java, Python, C/C++, JavaScript, HTML/CSS, x86 Assembly, R, Linux, Scala, TypeScript, Go
Frameworks & Tools: Git, AWS, Google Cloud Platform, PyTorch, Snowflake, Tensorflow, Keras, Django, React, Node.js, Flask, pandas, NumPy, Matplotlib, OpenCV, Scikit-learn, JUnit, Cucumber
Languages: English (Native), German (Fluent), Korean (Proficient), Spanish (Proficient)